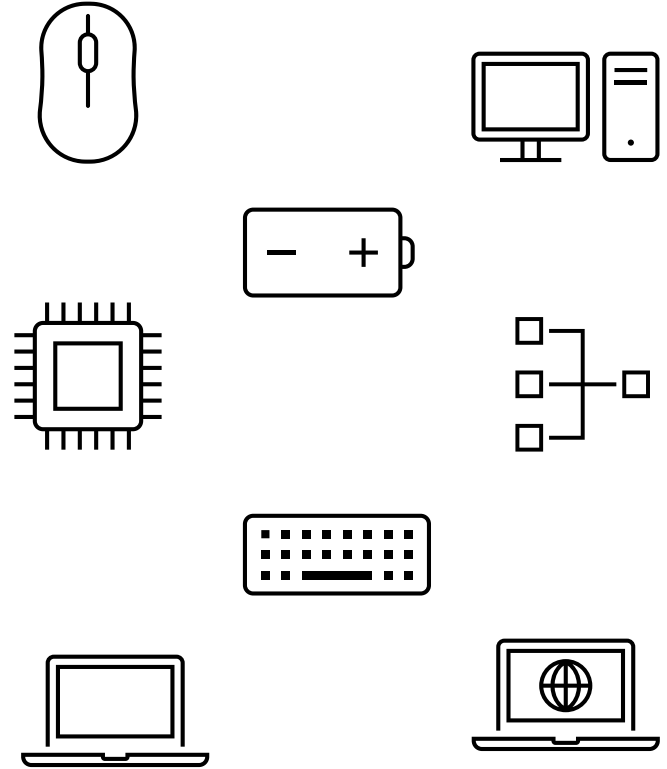
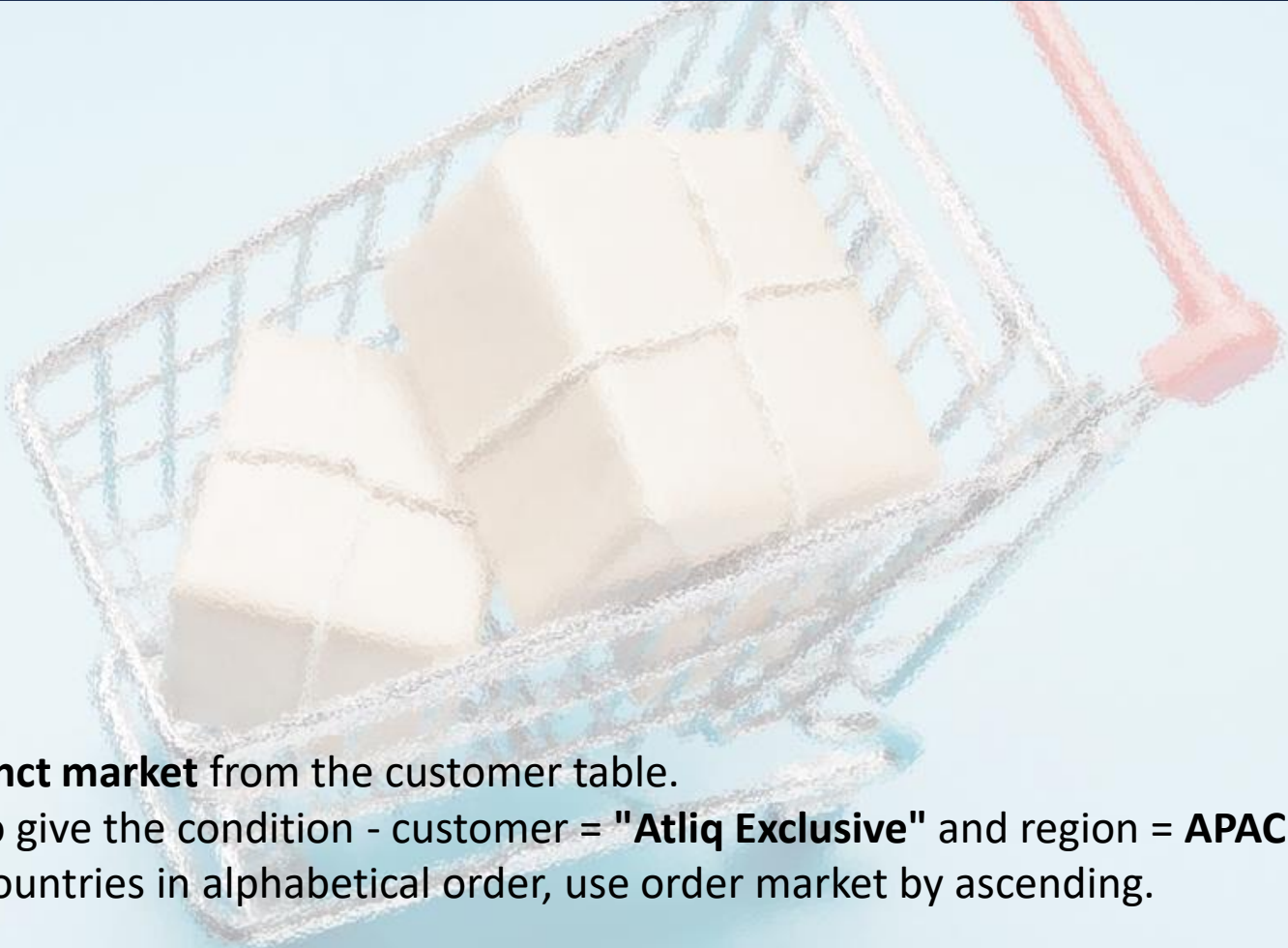


Consumer Goods Queries & Insights

SQL, Power BI, Power Point



Provide the list of markets in which customer "Atliq Exclusive" operates business in the APAC region.



	market
▶	Australia
	Bangladesh
	India
	Indonesia
	Japan
	Newzealand
	Philiphines
	South Korea

Explanation:

- We select all the **distinct market** from the customer table.
- We use where class to give the condition - customer = "**Atliq Exclusive**" and region = **APAC**.
- To display the list of countries in alphabetical order, use order market by ascending.

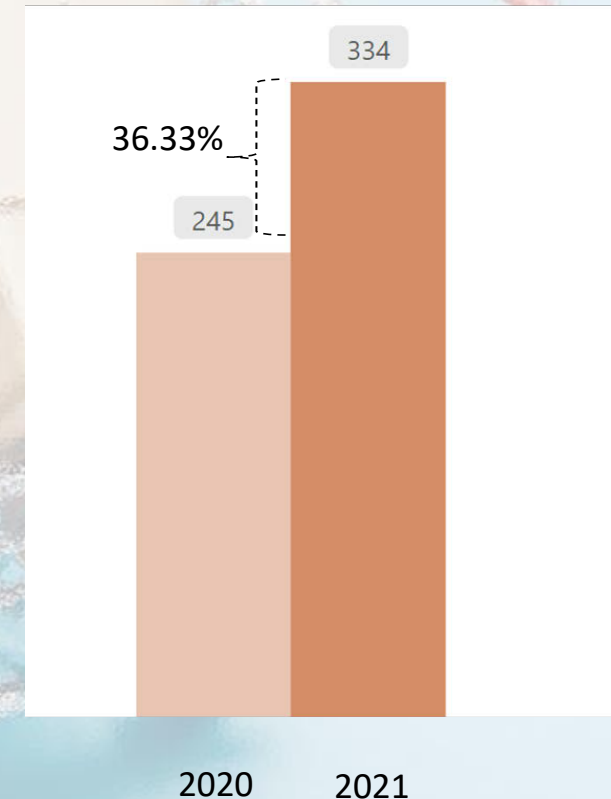
What is the percentage of unique product increase in 2021 vs. 2020?

The final output contains these fields : unique_products_2020, unique_products_2021, percentage_chg

	unique_products_2020	unique_products_2021	percentage_chg
►	245	334	36.33%

Explanation:

- From the sales table, we count the number of unique products for the fiscal year 2020 & 2021 separately using Case statements.
- With this result stored as a CTE named unique products, we find the percentage of unique products increase from 2020 to 2021.
- Round the results to two decimal and concat a percentage symbol for aesthetic reason.



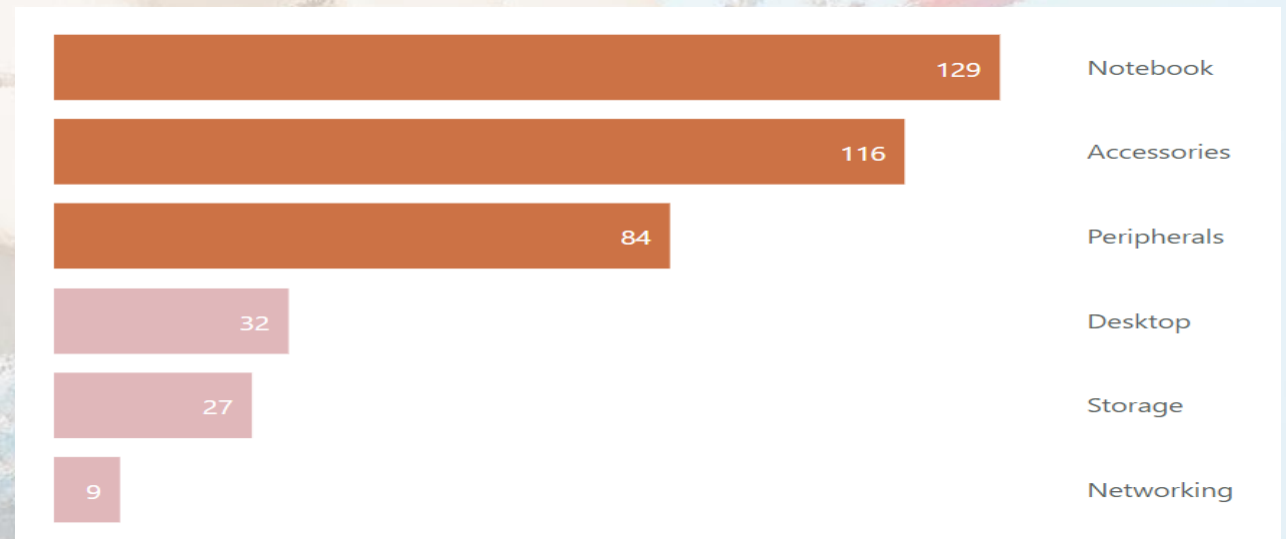
Provide a report with all the unique product counts for each segment.

The final output contains 2 fields - segment & product_count. Sort them in descending order of product counts.

	segment	product_count
▶	Notebook	129
	Accessories	116
	Peripherals	84
	Desktop	32
	Storage	27
	Networking	9

Explanation:

- From product table, we choose the column segment and count the distinct number of product as product count.
- Since we use an aggregate function, we have to use group by.
- The final result is ordered by the product count in a descending order



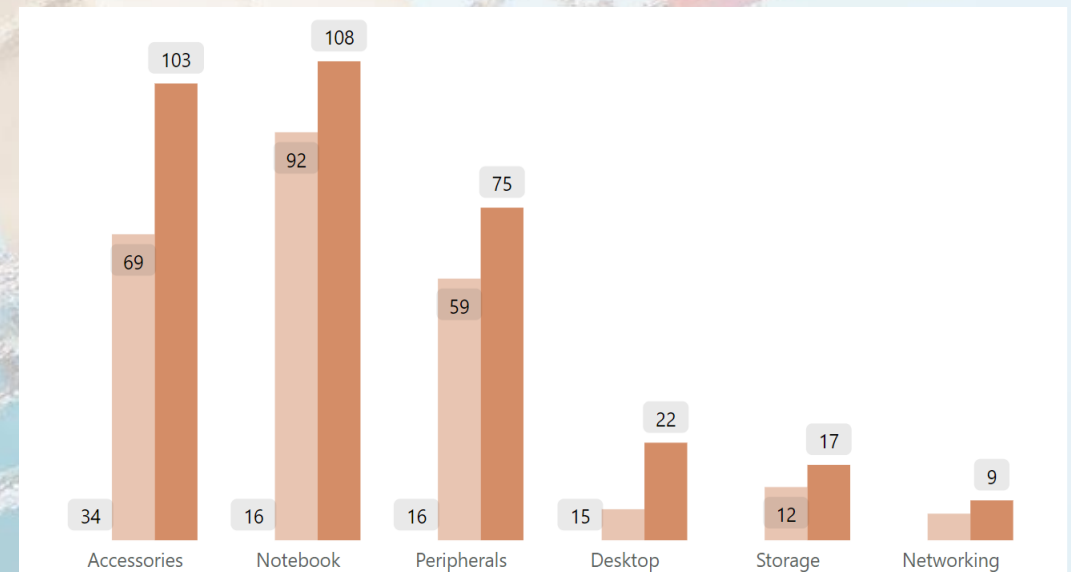
Follow-up: Which segment had the most increase in unique products in 2021 vs 2020?

The final output contains these fields:-- segment, product_count_2020, product_count_2021, difference

	segment	product_count_2020	product_count_2021	difference
►	Accessories	69	103	34
	Notebook	92	108	16
	Peripherals	59	75	16
	Desktop	7	22	15
	Storage	12	17	5
	Networking	6	9	3

Explanation:

- We find the unique product count for each fiscal year like before using CASE statements.
- Group by segment to get the same for each segment.
- using the above as a CTE, find the difference in the product count and display the result.



Get the products that have the highest and lowest manufacturing costs.

The final output should contain these fields: product_code, product, manufacturing_cost



	product_code	product	manufacturing_cost
▶	A6120110206	AQ HOME Allin1 Gen 2	\$ 240.54
	A2118150101	AQ Master wired x1 Ms	\$ 0.89

Explanation:

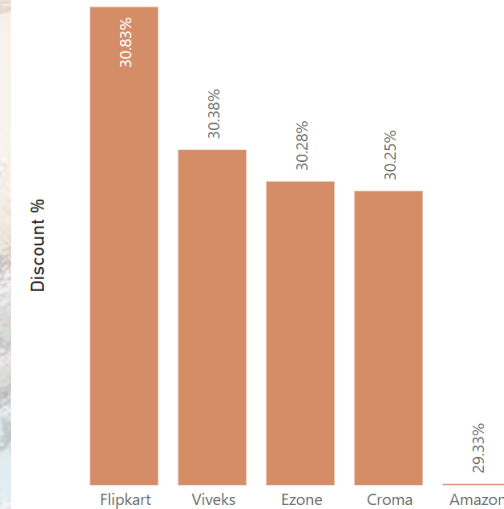
- We **join** the tables - manufacturing cost and products.
- Using separate **subqueries** to get the maximum and minimum manufacturing cost.
- This value is used in **where clause** to filter the result.

Generate a report which contains the top 5 customers who received an average high pre invoice discount pct for the fiscal year 2021 in the Indian market.

The final output contains these fields: customer_code, customer, average_discount_percentage

	customer_code	customer	average_discount_percentage
►	90002009	Flipkart	30.83 %
	90002006	Viveks	30.38 %
	90002003	Ezone	30.28 %
	90002002	Croma	30.25 %
	90002016	Amazon	29.33 %

Note : Axis starts at 29.33% to highlight the difference.



Explanation:

- We join the required tables and give filter conditions in where clause.
- Since we need to calculate an aggregate function, use group by.
- To get the top 5, order by descending and limit the results to 5 rows.

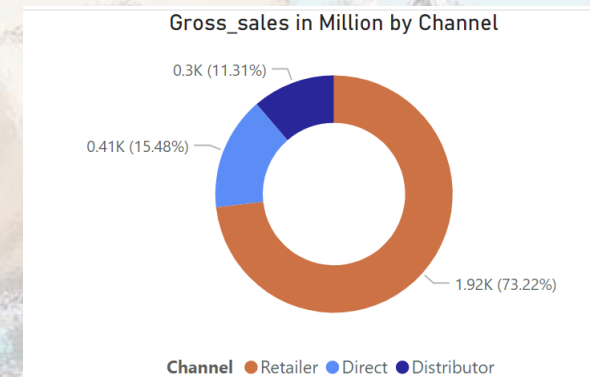
Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution?

The final output contains these fields: channel, gross_sales_mln, percentage

	channel	gross_sales_mln	percentage
▶	Retailer	1924.17	73.22 %
	Direct	406.69	15.48 %
	Distributor	297.18	11.31 %

Explanation:

- We compute the gross sales from gross price and sold quantity of the records from 2021, group it by channel to get gross sales million per channel for the fiscal year 2021.
- The obtained result is stored in a CTE.
- To find the percentage contribution of each channel we have to use window function over() on the gross sales million obtained in the above CTE.



Get the Top 3 products in each division that have a high sold quantity in the fiscal year 2021?

The final output contains these fields: division, product_code, product, total_sold_quantity, rank_order

	division	product_code	product	total_sold_quantity	rank_order
►	N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1
	N & S	A6818160202	AQ Pen Drive DRC	688003	2
	N & S	A6819160203	AQ Pen Drive DRC	676245	3
	P & A	A2319150302	AQ Gamers Ms	428498	1
	P & A	A2520150501	AQ Maxima Ms	419865	2
	P & A	A2520150504	AQ Maxima Ms	419471	3
	PC	A4218110202	AQ Digit	17434	1
	PC	A4319110306	AQ Velocity	17280	2
	PC	A4218110208	AQ Digit	17275	3

Explanation:

- Calculate the total sold quantity for the fiscal year 2021 per product.
- To find the top three products for each division, we give this result a rank using window function, DENSE_RANK where it is partitioned by division and ordered by sold quantity.
- Store the above result in a CTE.
- We get the final result where I filter the rank to display only those values where the rank order<=3.